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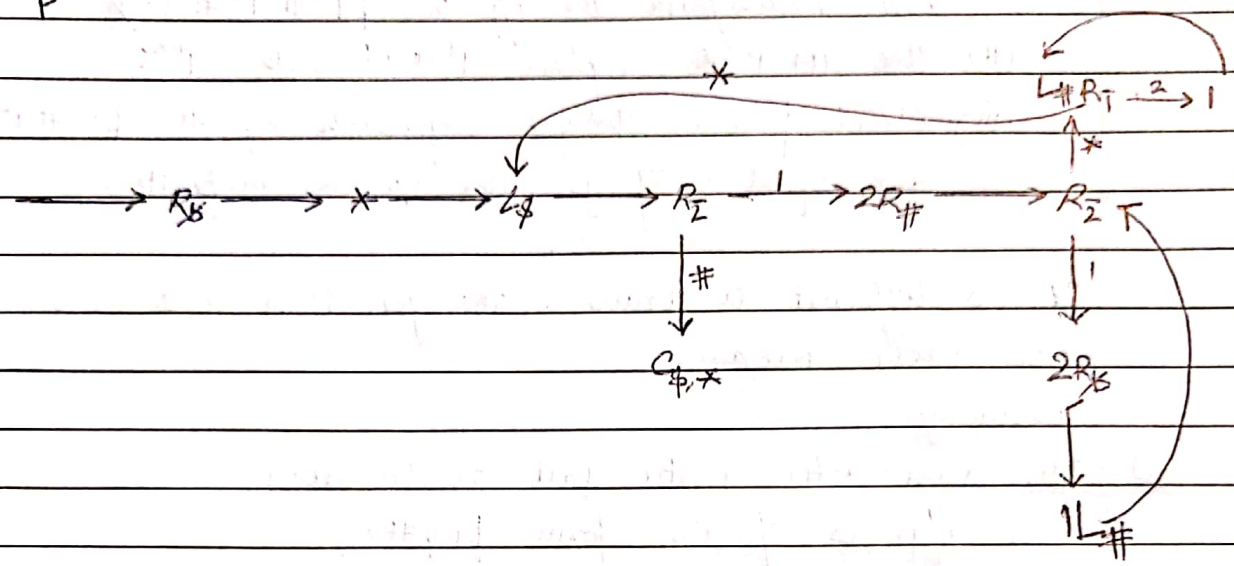
CSD 303: Quiz #6

A1: a) P



This machine calculates the unary product
If the input is of the form $\$1^m\#1^nB\ldots$
Then it loops through each m and within that each n (1 to m and 1 to n) and adds m×n 1s in this way at the end
The end of the input is signified by '*' and the machines uses a smaller TM $C_{a,b}$ to clear every thing on the tape between a and b, inclusive of b.
I use the symbol 2 to denote the position of the loop counter.

eg Input $\$111\#11111B\ldots$
Initial modification : $\$111\#11111*BB\ldots$
After processing : $\$222\#11111*1111111111111111B\ldots$
After using $C_{\$,*}$: $\$11111111111111B\ldots$

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b) We can construct an NTM that:

(i) Non-deterministically chooses p, q, s.t. $p, q \geq 1$ and transform the i/p to $\$1^{p^n}\#1^p\#1^q\not{b}\ldots$

(ii) Use TM P to replace $1^p\#1^q$ with $1^{p\times q}$

(iii) Check if the strings separated by # $\$1^n\#1^{p\times q}\not{b}\ldots$ are equal. If yes, the no. is composite.

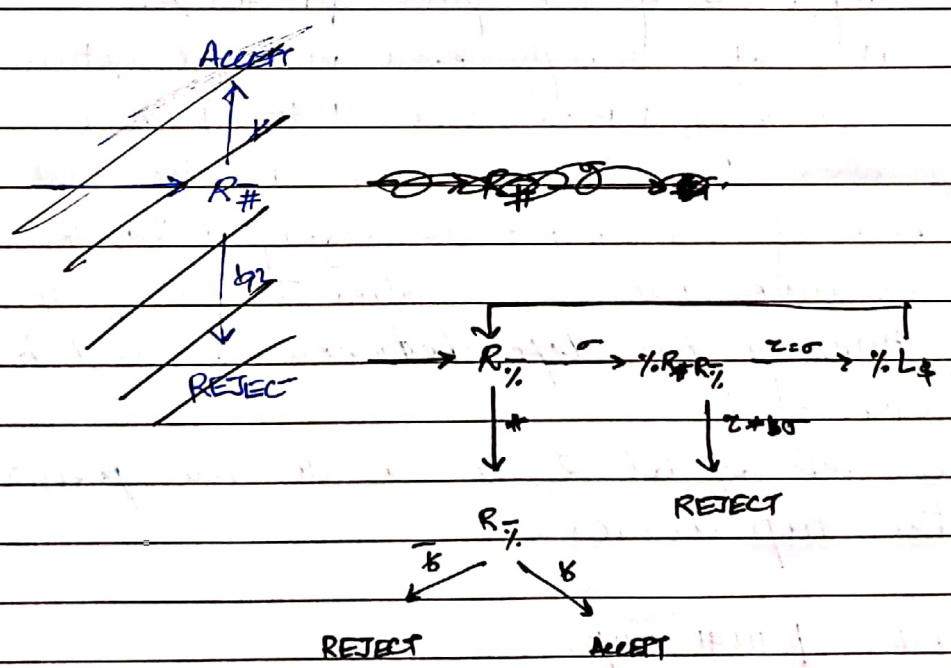
It is difficult to draw a TM for this but this is the basic idea.

Assuming,

The non-deterministic part to be done,

Our i/p is of the form $\$w_1\#w_2$

The comparison can be done as follows:



c) We can use a 2 tape TM

The 1st contains the i/p and is only read from. The 2nd represents the stack. Every time an element is pushed onto the stack, the second head would be moved R and the element would be written over a $\not{b}$ symbol.

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Similarly, for popping, the right most element on the 2nd tape replaced by $\not{b}$.

- $\delta(q, a, z) = (p, a)$
  - $\Rightarrow$ z is popped, a is pushed onto stack.
    i.e, rightmost symbol on 2nd tape is replaced.
- $\delta(q, a, z) = (p, \varepsilon)$
  - $\Rightarrow$ z is popped (and replaced with $\not{b}$)
- $\delta(q, a, \varepsilon) = (p, a)$
  - $\Rightarrow$ a is pushed onto stack.
    (replaces left most $\not{b}$)

~~A2: We can show that a BTM is equivalent to a single tape TM by showing it is equivalent to a 3 tape TM.~~
~~(We already know that a multitape TM ≡ Single tape TM)~~

~~To construct MTM...~~
~~(i) Initialize a 3 tape TM with the i/p str on every tape.~~
~~(ii) When the ith head moves to position j~~

A2 Let the states in BTM $B = \{q_1, q_2 \ldots q_k\}$. All transitions read a triple (r_1, r_2, r_3) from tape. Let us create a single tape TM M for the same. The states of M are $\{q_{11}, q_{12}, q_{13}, \ldots q_{k1}, q_{k2}, q_{k3}\}$. For a transition $q_i \to q_j$ in B, we add the foll. transitions to M:

- $q_{i1} \to q_{i2}$, on reading r_1 from tape head, moving head L/R
- $q_{i2} \to q_{i3}$, on reading r_2 from tape head, moving head L/R
- $q_{i3} \to q_{j1}$, on reading r_3 from tape head, moving head L/R

same dir" movement of BTM

If the accept state is q_e, the accept state of M is q_{e1}. The TM BM is like B but each cell expands to 3 cells, each storing as are symbol.

Hence, BTMs are equivalent to std. single-tape TM.

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A3. We know that Turing recognizable languages are recursively enumerable (r.e.). So, we need to show a language is Turing recognizable iff it can be enumerated by a TM

1) Let L be a Turing recognizable (t.r.) language. Let M be the TM that recognizes L. Using M, we can construct another TM E that enumerates L. Let $s_1, s_2, \ldots$ be the list of all strings in Σ^*.

We define E as:

(i) Ignore input
(ii) Write \$ on the first tape
(iii) Repeat for $i = 1, 2, 3 \ldots$
- Run M for i steps on each i/p.
- If any computations accept, write out corresponding s_i on first tape. Append o/p with #

If M accepts a string s_i it will eventually appear on 1st tape. Here, we never move the head of the 1st tape to the L. At all points, the tape consists of strings in L separated by #. When a string is written to the tape, a substr. will be present as string is written L → R.

$\Rightarrow$ TM E enumerates L (r.e.)

2) Let E be a TM that enumerates the lang L. We will construct a TM L that recognizes L. Let w be the input to M. We can construct M from E. Whenever E writes to its 1st tape, we also compare that o/p to the i/p string w. If w equals o/p, M accepts.
Clearly, M accepts all strings produced by E.
$\Rightarrow$ M recognizes L
$\therefore$ A lang is r.e. iff it can be enumerated by a TM.

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